RGSW based Universally Composable Oblivious Transfer

Group 22



Guide: Dr. Kunwar Singh

Members:







Anurag Goyal	Manmohan Prajapat	Sobhagya Singh Dewal
106119014	106119072	106119117

Introduction

We have constructed a secure method for transferring a message without knowing which one the receiver wants. The receiver also can't know the other messages. This is called 1-out-of-n Oblivious Transfer (OT) protocol.

We use a fully homomorphic encryption (FHE) scheme. It allows us to perform operations on ciphertexts in an untrusted environment to achieve the same results on the resulting plaintext.

The protocol is secure in the universal composability (UC) framework. Its post-quantum security comes from the RLWE hard problem. We achieved this by constructing a Ring-LWE based variant of the GSW FHE scheme.

Literature Survey

[3] Regev, Oded. (2005) Learning with Errors (LWE) is a lattice-based hard problem which is secure against quantum computers. It is the basis for several post-quantum encryption schemes.

[5] Gentry, Sahai, Waters (2013) Gentry et al. proposed an asymmetric encryption scheme based on the LWE hard problem. It shows fully homomorphic properties.

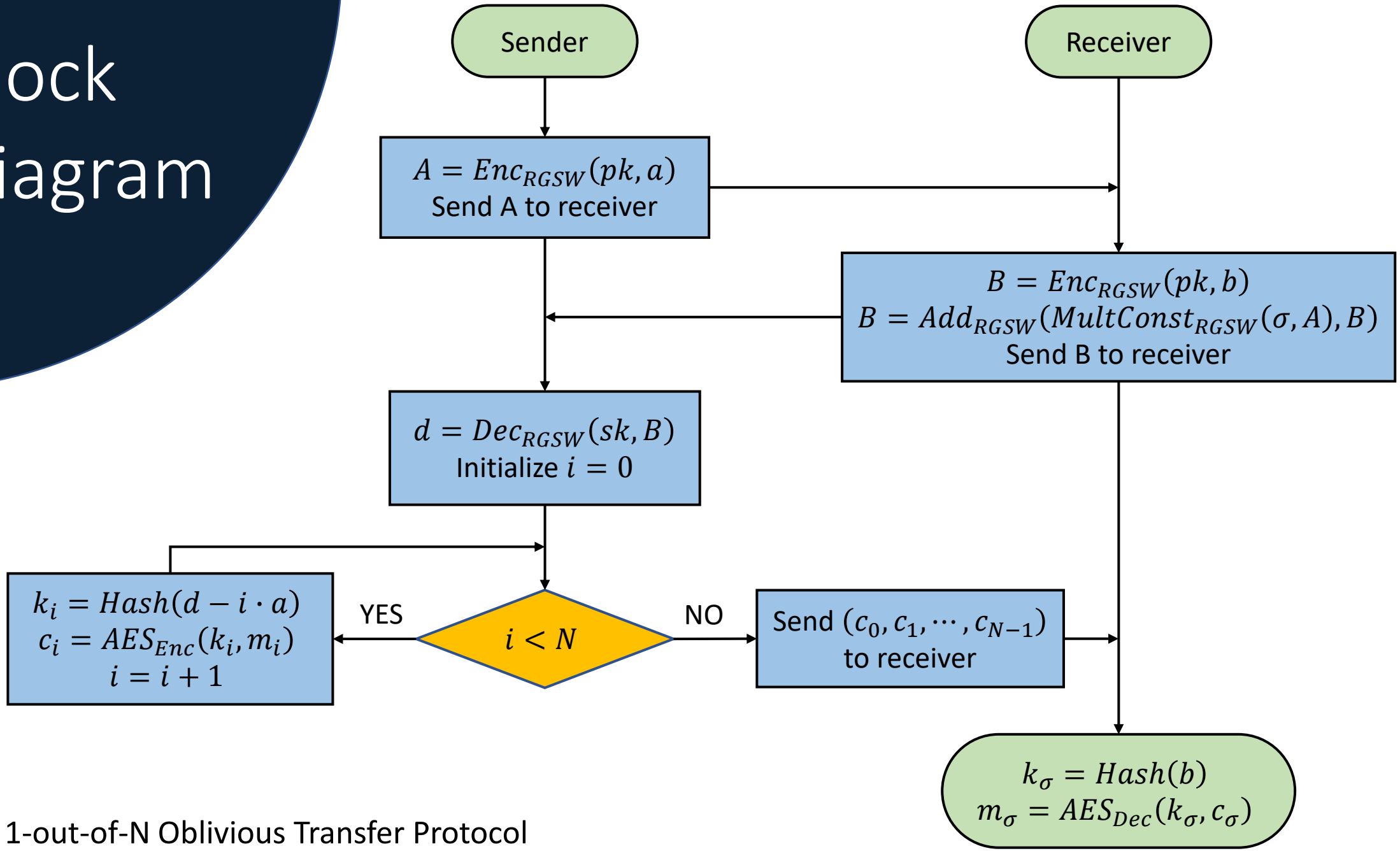
[6] Micciancio, Piekert (2012) The authors presented trapdoors for lattices for a special modulus. The use of this modulus makes it possible to design a generalized decryption algorithm for GSW.

[4] Piekert, Chris, Zachary (2019) Ring-LWE is a variation of LWE which utilizes the algebraic structure of polynomial rings which makes operations such as polynomial multiplication significantly faster. This problem is essential for constructing our RGSW scheme.

[1] Esmaeilzade, Saeid, Nasrollah, Ziba (2022) The authors proposed a generic framework for constructing oblivious transfer protocols which are universally composable. We have introduced post-quantum security to this.

Limitations of existing works	Objectives of our work
• Conventional lattice-based FHE schemes don't provide **full** homomorphism because of uncontrolled noise growth with each operation. • The existing LWE based FHE schemes suffer from huge public key and ciphertext sizes. Conversion to ring variants improve the protocol's efficiency without compromising on its security. • The GSW scheme achieves full homomorphism but there is room for improvement in its efficiency.	• To construct a ring variant of the GSW FHE scheme based on the Ring-LWE problem. • Achieving better efficiency than GSW in terms of key and ciphertext size. • To construct and implement a 1-out-of-n OT protocol using the homomorphic properties of RGSW. • To ensure the OT protocol is universally composable and post-quantum secure under the Ring-LWE assumption.

Block Diagram



1-out-of-N Oblivious Transfer Protocol

Algorithms

Polynomial Ring:

$R_q = \mathbb{Z}_q[x]/(x^n + 1)$ where q and n are powers of 2. Thus, R_q is a polynomial ring which has coefficients under modulo q. Let $q = 2^k$ and define $N = 2k$.

Ring variants of GSW methods:

$$BitDecomp_{RGSW}(a) = \left(\sum_{i=0}^{n-1} a_{i,0} \cdot x^i, \cdots, \sum_{i=0}^{n-1} a_{i,k-1} \cdot x^i \right) \in R_q^k$$

$$BitDecompInv_{RGSW}(\vec{a}) = \sum_{i=0}^{n-1} \left(\sum_{j=0}^{k-1} 2^j \cdot a_{i,j} \right) \in R_q$$

$$Flatten_{RGSW}(\vec{a}) = BitDecomp_{RGSW}\left(BitDecompInv_{RGSW}(\vec{a})\right) \in R_q^k$$

$$PowersOf2_{RGSW}(a) = \left(a, 2a, 2^{k-1}a\right) \in R_q^k$$

RGSW FHE Scheme

Setup: Let $\chi_{\mu,\sigma}$ be a gaussian distribution and I_N is an identity matrix of order N.

Secret Key Generation: Choose $t \in R_q$ uniformly at random and define $\vec{s} = (1, -t)$. The secret key is

$$\vec{v} = PowersOf2_{RGSW}(\vec{s})$$

Public Key Generation: Choose $b_1 \in R_q$ and $e \leftarrow \chi \in R_q$ uniformly at random where e is an error polynomial with small coefficients.

Assign $b_2 = b_1 \cdot t + e$. The public key is the matrix given by

$$A = \begin{bmatrix} b_2 & b_1 \end{bmatrix} \in R_q^{1\times 2}$$

$$A \cdot \vec{s} = e$$

Encryption: Choose a plaintext $z \in R_q$ and a uniformly random $\vec{r} \in R_q^N$ such that the coefficients of any member polynomial are $0/1$. The ciphertext is the matrix given by

$$C = Flatten_{RGSW}\left(z \cdot I_N + BitDecomp_{RGSW}(\vec{r} \cdot A)\right) \in R_q^{N\times N}$$

Decryption: Let $z_0, z_1 \cdots, z_{k-1}$ be such that $z = \sum_{i=0}^{n-1} z_j \cdot x^j \in R_q$.

Consider $\vec{b} = (b_0, b_1, \cdots, b_{k-1}) = \left(z + e_0, 2z + e_1, \cdots, 2^{k-1}z + e_{k-1}\right)$

Initialize $z = 0 \in R_q$
for $j = 0, 1, \cdots, n-1$:
    $z_j = 0 \in \mathbb{Z}_q$
    for $i = k-1, \cdots, 1, 0$:
        $z_j = z_j + 2^{k-1-i} \cdot \left\langle b_{i,j} - 2^i z_j \notin \left[-\frac{q}{4}, \frac{q}{4}\right) mod\ q \right\rangle mod\ q$
    $z = z + z_j \cdot x^j$

Where $b_{i,j} \in \mathbb{Z}_q$ denotes the coefficient of x^j in b_i and $\langle E \rangle = \begin{cases} 1, & E\ is\ true \\ 0, & E\ is\ false \end{cases}$

Homomorphic Operations

Scalar Multiplication: For some ciphertext $C \in R_q^{N \times N}$ and a scalar $\alpha \in \mathbb{Z}_q$

$$C_\alpha = Flatten_{RGSW}(Flatten_{RGSW}(\alpha \cdot I_N) \cdot C) \in R_q^{N \times N}$$

Addition: For ciphertexts C_1 and $C_2 \in R_q^{N \times N}$ encrypted with the same public key

$$C_+ = Flatten_{RGSW}(C_1 + C_2) \in R_q^{N \times N}$$

Implementation and Simulation

- The RGSW FHE scheme is implemented in **SageMath**, a free open-source mathematics software programming language.

- The OT protocol is implemented as two classes, **OTSender** and **OTReceiver**. An interface is developed for these classes to interact with each other.

- The OT protocol is simulated as a server-client model over an untrusted network.



Simulation - OT Protocol

SageMath 9.3 Shell

```
(sage-sh) meet1@LAPTOP-R2K8452O2:implementation$ sage server.py

[*] listening on port 3000

connection accepted

messages (visible to sender):

0. 551c6380eb8bd4dfc5cec934775805ee
1. 6cae9bdf78c4bbf34704f19e14cdfd5f
2. ecca5b93d5497840a306f7345b3b8a7b
3. fc2f74a598b062ab6a72e0ab24281dbe
4. 31ec32d9e3e379b7013057e43965ce06
5. 3ebf9dd2c35619be0d8b6b633f7f1d09

sharing parameters


sending a random encryption (A)

recieved homomorphic encryption of A (B)

sending symmetric AES ciphertexts

[*] connection closed

(sage-sh) meet1@LAPTOP-R2K8452O2:implementation$
```

SageMath 9.3 Shell

```
(sage-sh) meet1@LAPTOP-R2K8452O2:implementation$ sage client.py

connected to sender

received parameters (ring, public key)

enter your choice(s) (0 to 5): 3

recevied the random encryption (A)

sending homomorphic encryption of A (B)

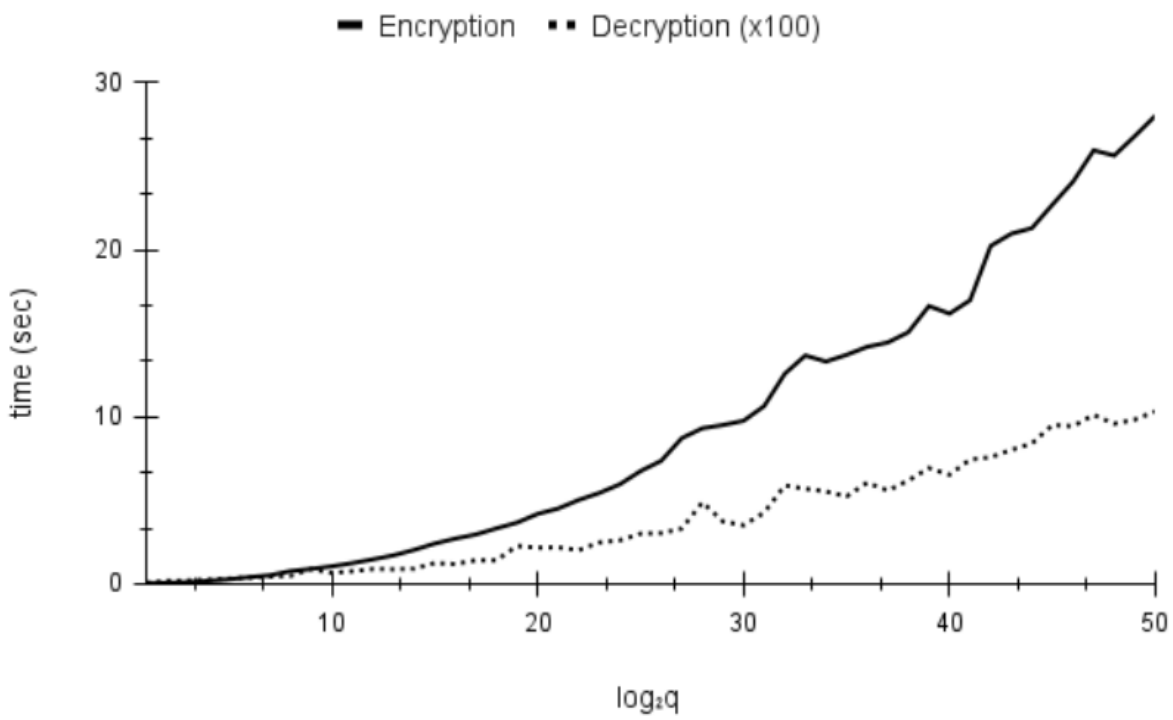
recevied symmetric AES ciphertexts:

0. af3533b0bb959d573932fde3ee7a9d0ddba128c9b97cbbf7e1
d67e100feac0e62f54790d40f0373f8f5905bd288400fb
1. d4fa2f13c02d8623e85c430046b6e0699e9ba7991d1a86b2d4
d0d9b8878e35cae7bbcd299ab6dee228054285ceaa8743
2. eb27f44182afc7580d9bfd4bf6717e3fa20fbde35a1a25d1b9
5e426d614961aabcc26759622a4174216cf1845b5668aa
3. 8e8ca353f2c560a7ab159fa62689bb39c1848ab23844853d1a
54527670508f57db40cb25d5b2cf7bdae3dd2f95eb57b8
4. a8254f5626da00af25b67e1d9d1e22714c68e9a7e0e2a05c79
0808d652e415cb23f58e2414d0b503f7c4000c22228e24
5. fedd5eec827bb6c4561c95fcd18f80b14d71e6245622114549
e25cf3e4cc04bbe09d5f805751485ff05e8b17652fa9ac

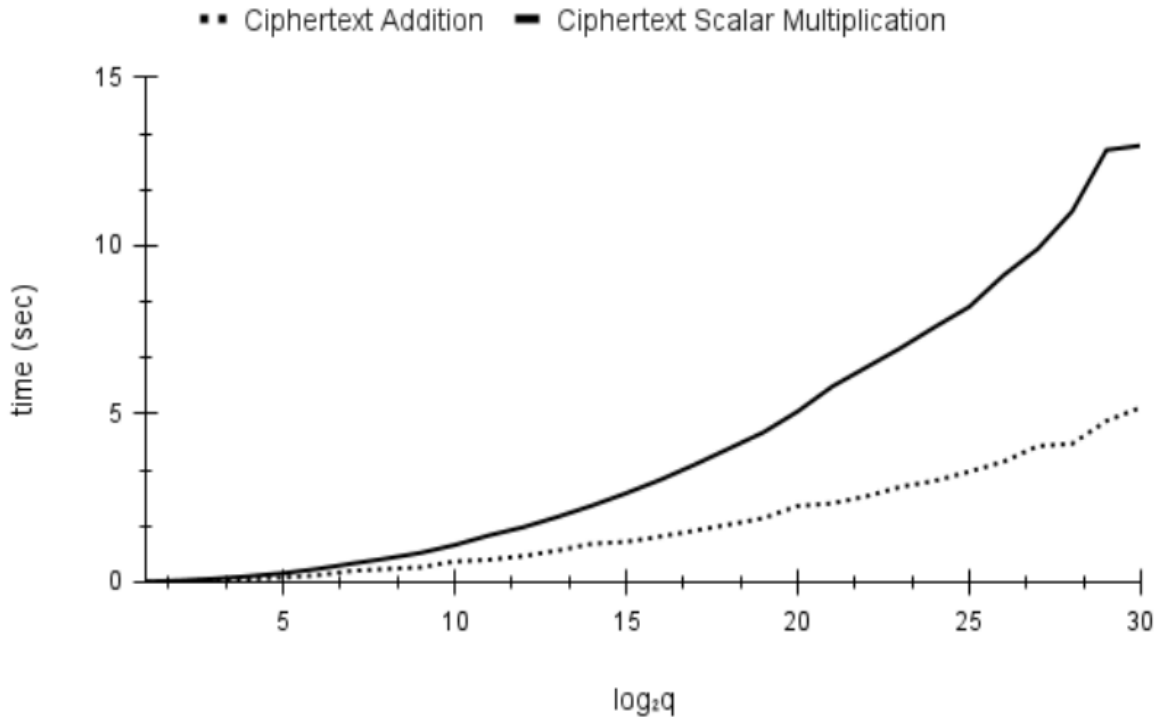
decrypted message:
3. fc2f74a598b062ab6a72e0ab24281dbe


[*] connection closed
```

Results - RGSW

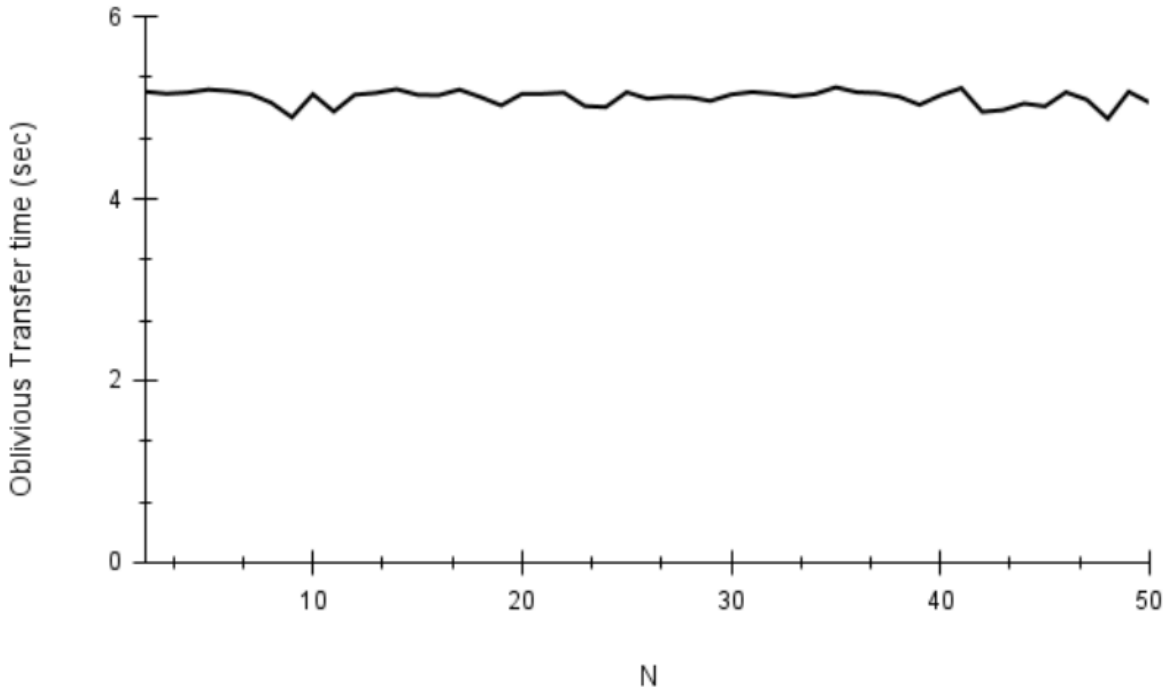


Encryption and Decryption time VS $log_2 q$ at n = 16

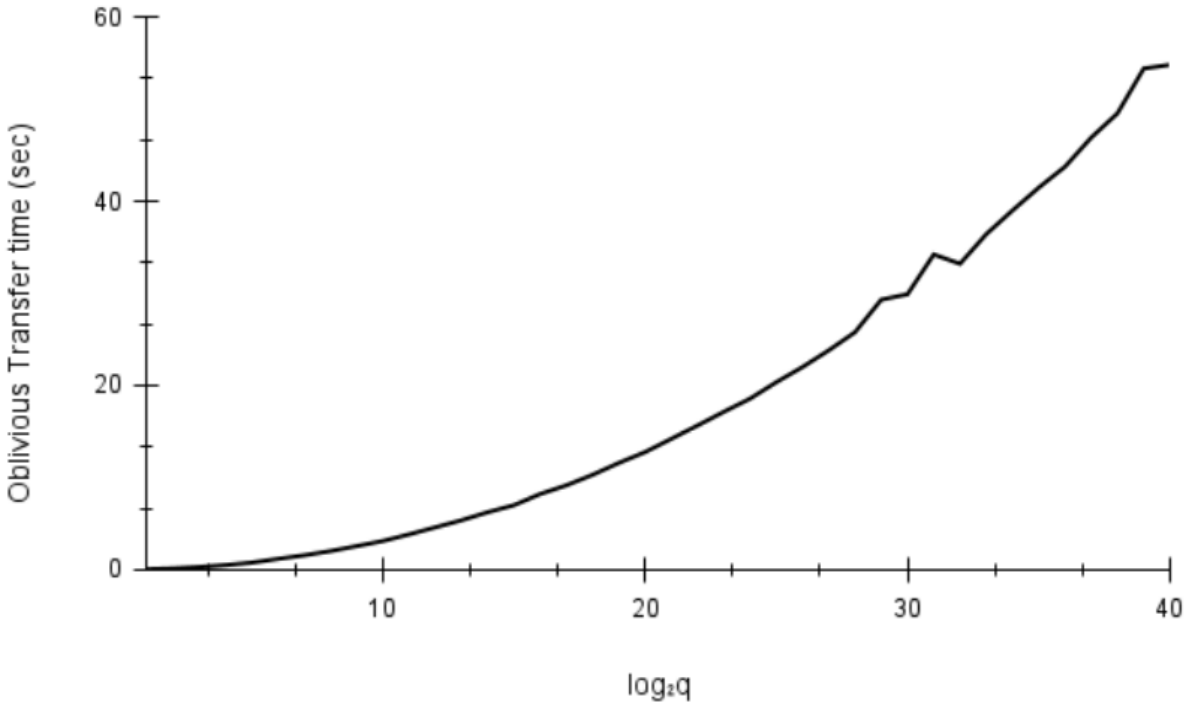


Add and MultConst time VS $log_2 q$ at n = 16

Results – OT Protocol



Oblivious transfer time VS N at n = 16, q = 8192



Oblivious transfer time VS $log_2 q$ at n = 16, N = 20

References

[1] Esmaeilzade, Saeid & Pakniat, Nasrollah & Eslami, Ziba. (2022).Ageneric construction to build simple oblivious transfer protocols from homomorphic encryption schemes. The Journal of Supercomputing. 78. 10.1007/s11227-021-03826-0.

[2] Canetti, R.. (2001). Universally Composable Security: A New Paradigm for Cryptographic Protocols. IACR Cryptology ePrintArchive. 2000. 136 - 145. 10.1109/SFCS.2001.959888.

[3] Regev, Oded. (2005). On Lattices, Learning with Errors, Random Linear Codes, and Cryptography. Journal of theACM (JACM). 56. 84-93. 10.1145/1568318.1568324.

[4] Peikert, Chris & Pepin, Zachary. (2019).Algebraically Structured LWE, Revisited. 10.1007/978- 3-030-36030-6_1.

[5] Gentry, C., Sahai,A., Waters, B. (2013). Homomorphic Encryption from Learning with Errors: Conceptually-Simpler, Asymptotically-Faster, Attribute-Based. In: Canetti, R., Garay, J.A. (eds) Advances in Cryptology – CRYPTO 2013. CRYPTO 2013. Lecture Notes in Computer Science, vol 8042. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-40041-4_5

[6] Micciancio, D., Peikert, C. (2012). Trapdoors for Lattices: Simpler, Tighter, Faster, Smaller. In: Pointcheval, D., Johansson, T. (eds)Advances in Cryptology – EUROCRYPT 2012. EUROCRYPT 2012. Lecture Notes in Computer Science, vol 7237. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-29011-4_41

[7] Chen, H., Dai, W., Kim, M., Song, Y. (2021). Efficient Homomorphic Conversion Between (Ring) LWE Ciphertexts. In: Sako, K., Tippenhauer, N.O. (eds)Applied Cryptography and Network Security.ACNS 2021. Lecture Notes in Computer Science, vol 12726. Springer, Cham. https://doi.org/10.1007/978-3-030-78372-3_18